7.

Content	Reactant		Products	
	x	y	*l*	m
Stoichiometric coefficient	2	3	4	1
No. of moles allowed to react	8	15	–	–
No. of moles of reactant reacted and product formed	8	12	16	4
No. of moles of un-reacted reactants and the product formed	–	3	16	4

Limiting reagent **: x**

Product formed **: 16 moles of *l* & 4 moles of m**

Amount of excess reactant **: 3 moles of y**

8.

$$\overset{+3}{As}_2\ \overset{-2}{S}_3 + H\overset{+5}{N}O_3 \rightarrow H_3\overset{+5}{As}O_4 + H_2\overset{+6}{S}O_4 + \overset{+2}{N}O$$

$2\times2e^-$ $3\times8e^-$ $3e^-$

$4e^- + 24e^-$

$28e^-$

Equate the total no. of electrons in the reactant side by cross multiplying,

$\Rightarrow$ $3\ As_2S_3 + 28\ HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$

Based on reactant side, balance the products

$\Rightarrow$ $3\ As_2S_3 + 28\ HNO_3 \rightarrow 6\ H_3AsO_4 + 9\ H_2SO_4 + 28\ NO$

Product side :36 hydrogen atoms & 88 oxygen atoms

Reactant side :28 hydrogen atoms & 84 oxygen atoms

Difference is 8 hydrogen atoms & 4 oxygen atoms

$\therefore$ Add 4 H_2O molecule on the reactant side.

Balanced equation is,

$3\ As_2S_3 + 28\ HNO_3 + 4\ H_2O \rightarrow 6\ H_3AsO_4 + 9\ H_2SO_4 + 28\ NO$

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